RUN · BETTER · V35 PASSED 100 / 100 LIVE FINAL 12:00

A buyer that negotiates on total cost, not unit price.

Rules only. Every claim cited from the data. Every decision logged with its cost breakdown. Practice mean 0.72 over ten cases, ten of ten deals, reproduced one hundred sessions in a row.

Hacker Noobs · Niko · Petra · Alex

github.com/aldfjla/MUCHackatonFinal · tag submission

01 · THE JOB

Close a term sheet by e-mail. Get scored on the whole cost.

SCORE

```
(C_open - C_deal) /
(C_open - C_best)
```

clipped to 0..1. Worse than the opening quote, a broken must-have, or no deal: 0.

WHAT C COUNTS

- price over the horizon, old price until the effective date
- payment days · freight · late delivery · tooling
- index-clause risk · quality commitments
- supplier health, in strategic categories

THE OPPONENT

- LLM supplier bots with hidden floors
- ten rounds per thread, 60 s per message
- a final offer that walks if pushed
- RFQs with three suppliers at once

live final · 5 waves × 8 min · one attempt per case · all teams at once

02 · THE PIPELINE

Four stages per message, all deterministic rules

• PREPARE

Cited evidence

Cost-driver indices, our purchase history, should-cost, scorecards, competing bids, contract notice periods. Each claim carries the IDs the arena verifies.

→

• READ

Sheets and status

Which asks the supplier refused or repriced, whether it is short of cash, whether it sent a final offer. Never the leaderboard or results.

→

• DECIDE

Price every package

Anchor at 0.80 of the ask, match the supplier's visible concession, send one to three equal-cost sheets that trade levers. A guard enforces must-haves on every sheet.

→

• CLOSE

Whole package or nothing

Accept when the package beats the opening and the best alternative in the pessimistic world. Take a passing final offer at once. Never walk while a supplier is alive.

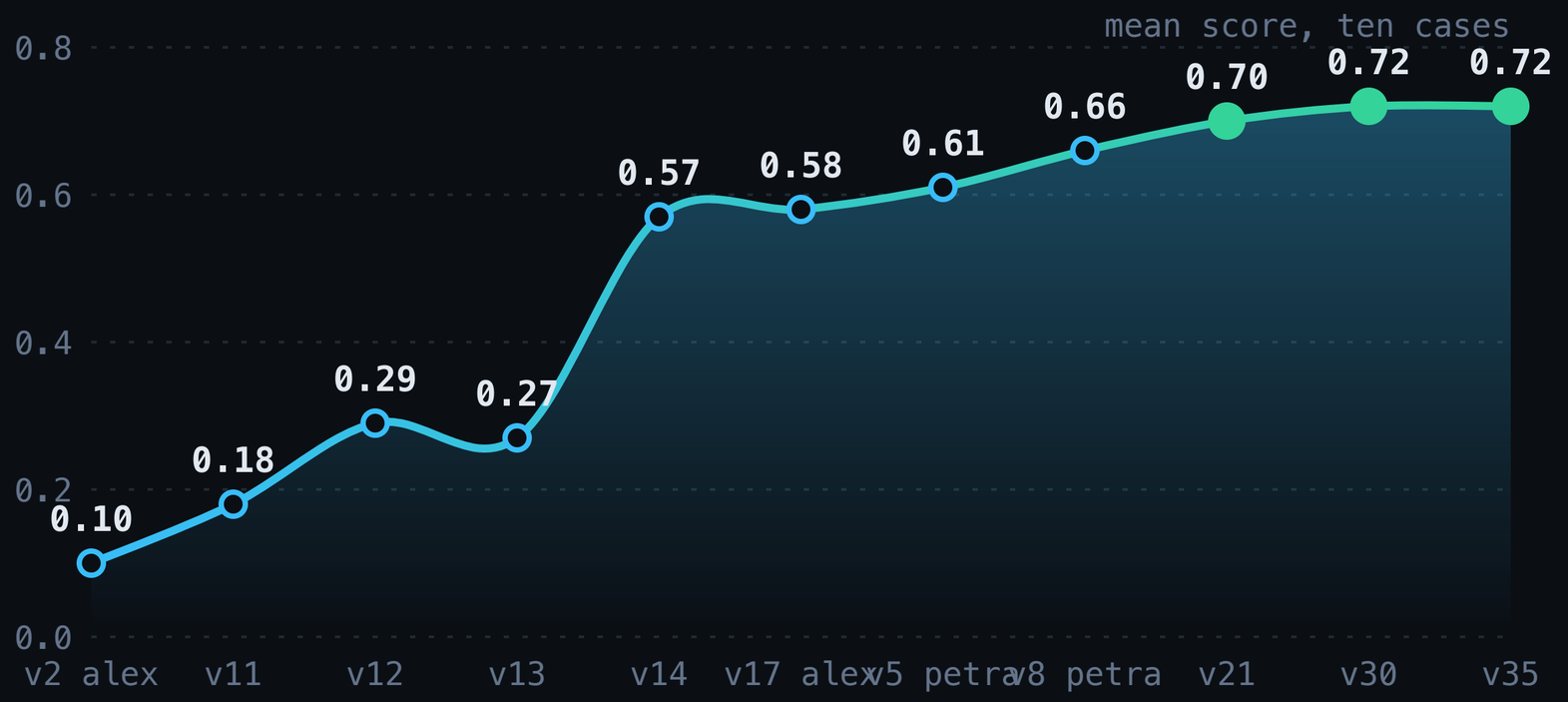
0 LLM CALLS PER NEGOTIATION

HEALTH BAND IN STRATEGIC CATEGORIES

EVERY DECISION LOGGED

03 · RUN HISTORY

24 hours, 403 logged sessions: 0.10 to 0.72



0.29 → 0.57

Match the supplier's visible concession, accept the final offer, 24-month contracts, cited evidence.

0.58 → 0.70

Price every package with the supplier-health term and an uncertainty band.

0.70 → 0.72

Latest effective date in every sheet (P01 0.69 → 0.88). No index sheet in the first message (P08 0.53 → 0.56).

source backlog.csv · first row per version and case · all three teammates' agents where they ran all ten cases

04 · SCENARIO WALL · V35

EUR 1.13 M saved of 1.56 M possible

P01
volute casing +9.5 %

0.88

50k saved · 7k left

P02
motor, copper clause

0.67

32k saved · 16k left

P03
converter renewal

0.92

311k saved · 26k left

P04
bearing renewal

0.86

2k saved · 0k left

P05
RFQ pump shaft

0.77

63k saved · 19k left

P07
fastener bundle

0.68

62k saved · 30k left

P08
1.4404 bar, alloy

0.56

99k saved · 77k left

P10
seal allocation cut

0.67

405k saved · 199k left

P11
bearing bracket quality

0.71

67k saved · 28k left

P18
1.4404 bar, Werdenfels

0.51

37k saved · 36k left

green ≥ 0.80 · blue ≥ 0.60 · amber below · "left" = what the best possible deal would still add · source: arena cost breakdowns, backlog.csv

05 · SUITE · 9 CHANGES · 2 PASSED · 7 FAILED

Same cases, one change at a time, the real arena is the verdict

change (general rules only)	mock	arena	verdict	what it taught us
latest effective date in every sheet	no signal	P01 +0.19 · 3/3	**PASSED**	the arena pays the old price until the date
no index-clause sheet in the first message	+0.00	P08 +0.03 · 3/3	**PASSED**	the first counter anchors the ladder
mirror the supplier's payment term	+0.03	mean −0.035	**FAILED**	payment days were not what moved P08
supplier-belief layer, one purposeful move	−0.08	mean −0.05	**FAILED**	a clarifying question hardened three suppliers
drop the clause sheet in health cases	−0.02	P02 −0.37 · 3/3	**FAILED**	the clause is what buys the price
ask the cost-minimising lead time	+0.05	P10 −0.02 · 3/3	**FAILED**	a full supplier hardens when pushed on time
scale the health term by 0.5	−0.02	P01 +0.05 · P02 −0.42	**FAILED**	the health charge is a cliff, not a scale
route index cases to the teammates' agent	−0.07	mean −0.04	**FAILED**	it ignores supplier health: P02 scores 0

mock = our simulator fitted on our own transcripts; agreed with the arena 4 of 8 times · every failed change stays in the codebase as a flag with its result in the comment

06 · REPLAY · CASE P01 · VOLUTE CASING +9.5 %

One thread, message by message

SUP	138.55 · start 2026-11-01
	opening ask, +9.5 % on the current 126.53
BUY	130.66 · 14 d · start 2027-04-01
	0.80 anchor; latest effective date the case allows; cites the 51-day notice vs the contract
SUP	138.40 April · or 135.30 January
	"we did not know that – a fair point"; prices the date itself
BUY	133.63 · April
	matches the supplier's visible concession
SUP	135.20 · April
BUY	133.77
	minimum step: keep the thread moving, never repeat a sheet
SUP	135.20 · "our calculation stays internal"
	held; resistance, not proof of a floor
BUY	133.91
SUP	134.90

VERDICT

0.88

EUR 50,350 of 56,964 possible. Five months at the old price are worth more than 2 EUR on the unit.

WHAT THE E-MAIL CITED

doc:20260911-70110-P01 (51 days notice) vs ctr:4600070110 · casting PPI 2025-08 to 2026-08 · labour cost index. All verified by the arena, none contradicted.

replay: backlog.csv row v30 - niko · case P01 · every session has one

07 · SUPPLIER MODEL · FROM 403 LOGGED SESSIONS

What we learned about the suppliers, and from where

DETERMINISTIC

100 of 100 sessions identical

Ten passes over ten cases reproduced every deal to the cent. One session per experiment is a perfect measurement; repeats are wasted quota.

FIXED CLOCK

Final offer at message 3 to 6

Each supplier sends it at the same message number every time; its size scales with our last step. Push past it and they walk.

EVIDENCE

Moves them only when it is new

"We did not know that, a fair point" on notice periods and energy costs. Labour indices: acknowledged in every case, acted on in none.

CLAUSE

The clause buys the price

A copper-exposed motor maker took 364.65 with an index clause at once and held 374 without one, three runs out of three.

CLIFF

Supplier health is a cliff

3.6 % under a supplier-specific price the health charge rose sixfold. Our band protects against it; scaling the term down walked us off the edge.

FLOORS

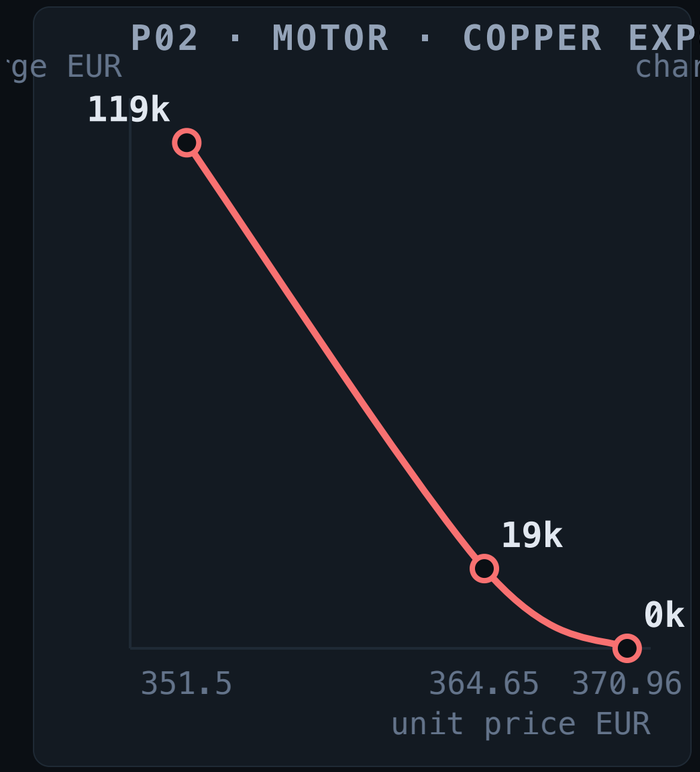
Some floors do not move

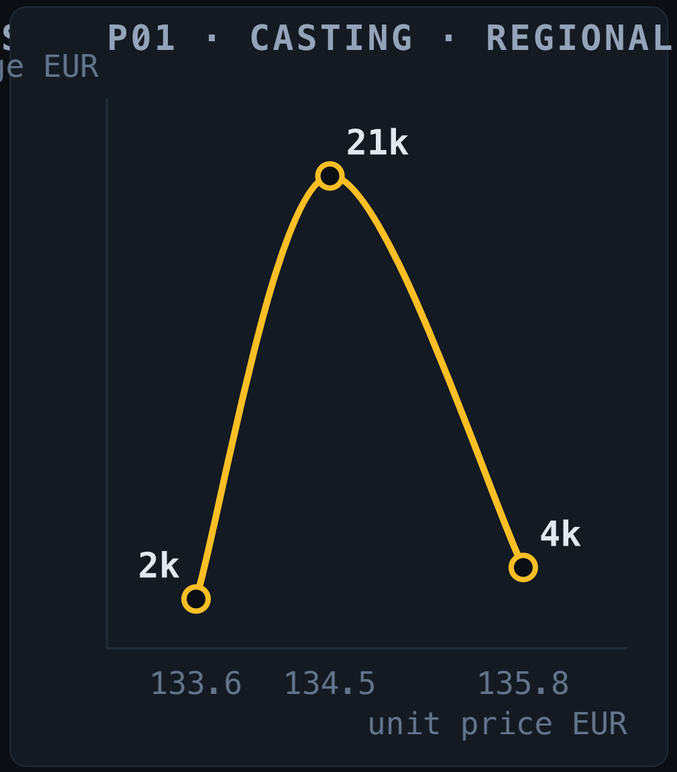
Raw-material and RFQ suppliers held against every anchor, timing and payment term we tried. There the best cost sits below any offer they ever made.

sources: our transcripts (agent/arena_kit/sim_cases/real_runs_*) · replays · the arena's cost breakdown per deal · LEARNINGS_v36.md

08 · MEASURED · SUPPLIER-HEALTH CHARGE VS CLOSING PRICE

The health cliff, measured on the arena





P02 · -3.6 % PRICE · ×6 CHARGE

The deal at 351.5 cost more in total than the deal at 364.65 and scored 0.25 instead of 0.67. Three sessions out of three.

WHAT THE AGENT DOES WITH IT

Estimates each supplier's full cost from the should-cost table, prices offers inside an uncertainty band, accepts only packages that pass in the pessimistic world. The lower edge of the band is the cliff.

measured points only · arena cost breakdowns, sessions v30, v36, v37 · P01's 2,242 is a January-start deal; the other two are April starts

09 · TRANSFER

Would it work on a real supplier?

CARRIES OVER

- Buying on total cost over the horizon, not unit price
- Every claim cited from our own data, so a buyer can stand behind the e-mail
- Package trades: payment, contract length, effective date, index clause
- An explicit supplier-health limit in strategic categories
- A log of why each offer was made and why each deal was accepted

DOES NOT, YET

- Real suppliers are not deterministic and do not announce a final offer
- The agent reads structure, not text: an objection only registers when it changes a sheet
- Fixed-clock timing rules would be wrong against a human
- No memory across negotiations with the same supplier

NEXT STAGE A language model that turns an objection into a structured hypothesis (what they value, what they refuse, what is uncertain), with the arithmetic and the accept checks staying deterministic. We built the deterministic half and tested it; the interpretation half is the open work.

One change at a time, judged by the real arena

90

tests: the concurrent play loop and every flag's on and off path

METHOD

Candidate vs unchanged baseline on identical cases; promote only what repeats; rejected ideas stay as default-off flags with their result in the comment. A fitted mock arena for crash and regression checks, never for tuning: it is in-sample and text-blind.

EXPLAINABILITY

Every decision writes its rule, the package cost breakdown, the health band and the concession history into the result. Every e-mail cites only facts the arena can verify; zero contradicted claims across all sessions.

backlog.csv · TIMELINE.md · LEARNINGS_v36.md · COMPARISON_better_vs_merged.md · agent/arena_kit/sim_cases/real_runs_*

RUN COMPLETE · IN ONE LINE

A buyer that prices the whole package, cites what it knows, and closes when the numbers say so.

0.72
practice mean · ten cases

10 / 10
deals · every run

100 / 100
sessions reproduced

1.13 M
EUR saved · of 1.56 M possible

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questions welcome